

Estate.ly

A Modern Real Estate Marketplace and Property Management Platform

1. Introduction

1.1 Motivation

The real estate industry has increasingly shifted toward digital platforms, where buyers, tenants, and investors search for properties online. However, many real estate websites still rely on fragmented listings, limited search capabilities, and outdated interfaces that make property discovery difficult and time-consuming. As a result, users often need to browse multiple platforms to find accurate property information and compare available options.

At the same time, property owners, developers, and real estate agents face challenges in efficiently managing property listings and reaching the right audience. These limitations highlight the need for a modern digital solution that simplifies property discovery while improving listing management.

Estate.ly was developed to address these challenges by providing a centralized real estate marketplace that integrates advanced search tools, intelligent filtering, AI-powered insights, and modern user experience design. The platform aims to connect buyers, tenants, property owners, developers, and agents within a single system, creating a more efficient and transparent real estate ecosystem.

1.2 Problem Statement

Despite the growth of digital marketplaces, the real estate sector still faces several challenges.

Despite the presence of several real estate websites, users still encounter various challenges when searching for properties or managing listings.

Limited Property Discovery

Many platforms provide only basic search features, making it difficult for users to filter properties based on precise criteria such as:

- price range
- location
- property type
- property size
- amenities

This makes it difficult for users to find properties that match their exact preferences.

Fragmented Property Listings and Information

Property information and listings are often scattered across multiple websites, agencies, and offline sources.

This fragmentation makes it difficult for buyers to compare listings and make informed decisions effectively.

Inefficient Listing Management

Real estate agents and property owners often lack centralized tools to:

- manage property listings
- update property details
- track inquiries
- analyze listing performance

Lack of Intelligent Recommendations

Most property platforms do not provide intelligent recommendations or AI-powered price insights that help buyers make informed decisions.

Poor User Experience

Some real estate websites have outdated interfaces, slow loading times, and complicated navigation structures that negatively affect user engagement.

Limited Market Transparency

Buyers and tenants often struggle to evaluate whether a property price is reasonable due to limited access to historical pricing data and market insights.

Proposed Solution

EstateLy addresses these challenges by introducing a modern digital platform that provides:

- A centralized property marketplace
- Advanced property search and filtering
- Interactive map-based property exploration
- AI-powered property price estimation
- Personalized property recommendations
- Agent and property management dashboards
- A powerful dashboard for agents and administrators

Through these features, EstateLy improves the overall efficiency, transparency, and usability of the real estate market for both buyers and property professionals.

1.3 Goals of the Project

The primary goal of EstateLy is to develop a modern real estate platform that simplifies property discovery and improves property management for both users and real estate professionals.

The platform aims to provide a centralized digital environment where buyers, tenants, property owners, developers, and agents can interact efficiently while accessing reliable real estate information.

Key Objectives

Improve Property Discovery

Enable users to easily search and explore properties using advanced filtering tools and intuitive navigation.

Enhance User Experience

Provide a modern, responsive, and user-friendly interface that allows users to browse properties smoothly across desktop and mobile devices.

Support Property Management

Provide tools that allow real estate agents, developers, and property owners to efficiently create and manage property listings.

Improve Decision Making

Provide intelligent insights such as price estimation and personalized recommendations that help users evaluate properties more effectively.

Provide Automated User Assistance

Integrate an intelligent chatbot that assists users by answering questions and guiding them through the property search process.

Ensure Platform Scalability and Security

Develop a reliable and scalable system capable of supporting a growing number of users while maintaining strong security and data protection standards.

1.4 Existing Real Estate Platforms

Several real estate platforms already operate in the Egyptian market.

Nawy

Nawy is a digital real estate brokerage platform that connects buyers with developers and agents.

It focuses mainly on connecting buyers with real estate developers and new projects.

While it provides useful developer-focused listings, it may offer limited flexibility for independent property owners and smaller real estate agencies.

Property Finder

Property Finder is one of the largest real estate marketplaces operating across the Middle East.

It provides:

- property listings
- search filters
- agent profiles

However, the platform lacks advanced AI-powered insights such as price prediction or personalized recommendations.

Bayut Egypt

Bayut offers property listings for residential and commercial properties across multiple regions.

Although it provides a strong property database and map-based browsing, the platform does not include AI-driven property analysis or recommendation systems.

Egyptian Government Real Estate Platform

This platform focuses on government-approved projects and real estate transparency.

However, it primarily serves as an informational platform and lacks advanced search, personalization, and agent management capabilities.

1.5 Feature Matrix

Feature	Nawy	Property Finder	Bayut Egypt	Egyptian Official Real Estate Platform	Estatefy
Property Listing System	✓	✓	✓	✓	✓
Advanced Search Filters	✓	×	×	✓	✓
AI-based Price Estimation	×	×	×	×	✓
Interactive Map View	✓	×	✓	✓	✓
Data Security	✓	✓	✓	✓	✓
Customer Support / Chatbot	×	×	×	×	✓
User Experience (UI/UX)	Advanced	Moderate	Moderate	Moderate	Advanced
Integration with Agents / Brokers	✓	✓	✓	×	✓
AI/ML Personalization	×	×	×	×	✓

1.6 Actors of the System

Actors represent the different types of users who interact with the Estatefy platform.

Property Buyers

Buyers are users who browse the platform to search for properties that they intend to purchase.

Buyer Activities

- Searching for properties
- Filtering listings

- Viewing property details
 - Comparing property options
 - Contacting agents
 - Scheduling property visits
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Tenants

Tenants use the platform to find rental properties such as apartments or houses.

Tenant Activities

- Searching for rental properties
 - Comparing rental prices
 - Viewing property images and details
 - Contacting property owners or agents
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Property Owners

Property owners are individuals who want to sell or rent their properties through the platform.

Owner Activities

- Creating property listings
 - Uploading images and property details
 - Managing listing status
 - Responding to user inquiries
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Real Estate Developers

Real estate developers are companies that build and sell residential or commercial projects such as compounds, residential communities, and commercial buildings.

Examples of developers in the Egyptian real estate market include companies such as:

- Palm Hills Development
- Mountain View
- Emaar Misr
- Talaat Moustafa Group
- Hassan Allam Properties

These developers often manage **large-scale real estate projects** that include multiple units and properties.

Developer Activities

- Publishing new real estate projects
- Listing multiple units within a project
- Uploading project details, images, and master plans
- Managing available units and pricing
- Responding to buyer inquiries
- Promoting new developments and investment opportunities

The platform allows developers to showcase their projects to a wide audience of potential buyers and investors.

Real Estate Agents

Agents represent clients and manage property listings on their behalf.

Agent Activities

- Managing multiple property listings
 - Communicating with potential buyers
 - Updating property information
 - Scheduling property visits
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System Administrators

Administrators manage the overall platform and ensure the system operates efficiently.

Admin Activities

- Managing users and agents
 - Approving property listing
 - Monitoring platform performance
 - Generating reports and analytics
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1.7 Software Development Methodology — Agile

The development of Estatefy follows the **Agile Software Development Methodology**, which is an iterative and flexible approach to building software systems.

Agile allows the development team to build the system in small iterations while continuously improving the platform based on feedback.

Agile focuses on continuous improvement, collaboration, and frequent delivery of functional components.

Why Agile Was Selected

Flexibility

Real estate platforms must evolve continuously based on market changes and user feedback.

Incremental Development

Core modules such as:

- user authentication
- property listing management
- property search functionality
- admin dashboard

were developed incrementally.

Each module is completed and tested before moving to the next.

Continuous Feedback

Regular reviews allow supervisors and potential users to provide feedback throughout development.

Early Issue Detection

Frequent testing helps detect performance or usability problems early.

Stakeholder Collaboration

Agile promotes collaboration between developers, designers, and stakeholders.

Agile Development Process

Sprint Planning

Development work is divided into short cycles called **sprints**, usually lasting **two weeks**.

Example sprint tasks:

- implementing property search functionality
- developing property listing pages
- building user authentication system

Daily Standup Meetings

Short daily meetings where team members discuss:

- progress on tasks
- challenges encountered
- planned work for the day

Sprint Review

At the end of each sprint, completed features are demonstrated to stakeholders and feedback is collected.

Sprint Retrospective

The development team evaluates the development process and identifies improvements for future iterations.

Benefits of Agile for Estatefy

Using Agile methodology provides several advantages:

- faster development cycles
- continuous feature improvement
- early bug detection
- higher software quality
- better collaboration between developers and stakeholders

This methodology ensures that Estatefy evolves continuously while maintaining a high-quality user experience.

1.8 System Scope

The Estately platform focuses on providing an online marketplace for property discovery and management. The system includes property listings, search tools, recommendations, chatbot assistance, and dashboards for agents and administrators. Financial transactions and legal property ownership verification are outside the scope of the system.

2. Requirements Analysis

2.1.1 [Survey](#) and Questionnaires

To better understand the needs of property buyers and sellers, a survey was conducted.

[Survey](#) details:

- Tool: Google Forms
- Participants: 120+ respondents
- Duration: 7 days
- Question types: multiple choice and rating scale

The [Survey](#) focused on:

- property search behavior
 - decision factors
 - challenges in finding properties
 - expectations from real estate platforms
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2.2 Functional Requirements

User Management

- Users can register and log in.
- Role-based access control is enforced.

Property Listings

- Agents can add, edit, and delete property listings.
- Listings include images, descriptions, and location data.

Property Search

- Users can search properties by:
 - location
 - price range
 - property type
 - area size
 - number of rooms

Property Details

- Users can view detailed property information.
- Image galleries and property features are displayed.

Appointment Scheduling

- Users can request property visits.

Admin Management

- Admins can manage users and property listings.
 - Admins can approve or reject listings.
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2.3 Non-Functional Requirements

Performance

- Search results should load within **2 seconds**.

Security

- HTTPS encryption and secure authentication.

Scalability

- The system must support **10,000 concurrent users**.

Usability

- Simple and intuitive user interface.

Reliability

- Platform uptime should reach **99.9% availability**.

Compatibility

- The website should work on all modern browsers and mobile devices
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2.4 Detailed Features of Estately

Estately includes several powerful features designed to improve the real estate discovery and management process.

Property Listing System

Agents and property owners can create detailed property listings including:

- property title
- description
- price
- location
- property type
- number of rooms
- property size
- amenities
- image galleries

Listings can be updated or removed at any time.

Advanced Search and Filtering

Users can search for properties using filters such as:

- property type
- price range
- location
- number of bedrooms
- property size
- available amenities

Interactive Map-Based Search

The platform includes a map view that allows users to explore properties geographically.

Users can:

- search properties within a specific area
- view nearby amenities
- identify property locations visually

AI-Based Property Price Estimation

Estatefy includes an AI-driven price estimation feature that analyzes market data to estimate the fair value of properties.

This helps users determine whether a property is overpriced or underpriced.

Personalized Property Recommendations

The system analyzes user preferences and browsing behavior to recommend properties that match their interests.

This helps users discover relevant properties more quickly and improves the overall search experience.

Property Visit Scheduling

Users can request appointments to visit properties.

Agents can confirm or reschedule appointments through the dashboard.

AI Chatbot Assistance

Estatefy includes an **AI-powered chatbot** designed to assist users while they interact with the platform.

The chatbot provides instant support by answering user questions and guiding them through the property search process.

The chatbot can help users by:

- answering frequently asked questions about properties and listings
- guiding users in searching for properties using filters
- providing quick information about property details and locations
- helping users contact agents or request property visits
- assisting users in navigating different sections of the platform

This feature improves user engagement and ensures that users receive immediate assistance while exploring the website.

Favorite / Saved Properties

Estatefy allows users to save properties they are interested in by adding them to a **Favorites list**.

This feature helps users easily keep track of properties they want to review later without searching for them again.

Users can:

- add properties to their favorites list
- remove properties from the favorites list
- quickly access saved properties from their user profile
- compare saved properties before making a decision

The Favorites feature improves the user experience by allowing property seekers to organize and manage the listings they find most relevant during their search process.

Admin Dashboard

Administrators can:

- manage users
- approve property listings
- monitor system activity
- generate reports

2.5 Technology Stack

The development of the Estatefy platform relies on a modern and scalable technology stack that supports efficient backend processing, secure data management, and a responsive user interface. The selected technologies were chosen for their reliability, performance, and strong ecosystem support.

Backend Framework — ASP.NET MVC (.NET)

The backend of the system is developed using **ASP.NET MVC**, a framework within the .NET ecosystem that follows the **Model–View–Controller architectural pattern**.

ASP.NET MVC separates the application into three main components:

- **Model** – Represents the data and business logic of the application.
- **View** – Responsible for presenting the user interface.
- **Controller** – Handles user requests and communicates between the model and the view.

This architecture improves:

- code organization
- maintainability
- scalability
- testability

ASP.NET MVC also provides built-in features such as routing, authentication support, and strong integration with database technologies.

Database Management System — Microsoft SQL Server

The platform uses **Microsoft SQL Server (MSSQL)** as the primary relational database management system.

SQL Server is used to store and manage all application data, including:

- user accounts
- property listings
- property images
- property visit requests
- saved properties

- chatbot interaction data

SQL Server provides high performance, strong security features, and reliable data management capabilities suitable for large-scale web applications.

Object Relational Mapping — Entity Framework

The system uses **Entity Framework (EF)** as an Object Relational Mapping (ORM) tool.

Entity Framework allows developers to interact with the database using **C# objects instead of writing raw SQL queries**.

Key advantages include:

- simplified database operations
- reduced development time
- automatic mapping between database tables and application models
- support for migrations and schema updates

This improves development efficiency and reduces the likelihood of database-related errors.

Frontend Technologies

The user interface of the Estatefy platform is built using modern web technologies including:

- **HTML5** – for structuring web pages
- **CSS3** – for styling and layout design
- **JavaScript** – for dynamic user interaction

These technologies ensure that the platform provides a responsive and interactive user experience across different devices and screen sizes.

Version Control System — Git

The project uses **Git** for version control to track changes in the codebase and manage collaborative development.

Git allows developers to:

- track modifications to the source code
- collaborate efficiently
- maintain code history
- revert changes when necessary

Development Tools

Several development tools were used during the implementation of the system:

- **Visual Studio** – primary development environment for .NET development
- **SQL Server Management Studio (SSMS)** – for database design and management
- **GitHub** – for source code hosting and collaboration

These tools support efficient development, debugging, and deployment of the platform.

2.6 System Architecture

The Estately platform is designed using a combination of **Onion Architecture** and the **Model–View–Controller (MVC) pattern**.

This architecture ensures high maintainability, scalability, and separation of concerns within the system.

While **MVC** organizes the interaction between the user interface and the application logic, **Onion Architecture** defines the internal structure of the system by separating responsibilities into multiple layers.

The main principle of Onion Architecture is that **dependencies flow inward**, meaning that the inner layers of the system remain independent from external frameworks and technologies.

Onion Architecture Layers

The Estately platform is divided into several layers, each with a specific responsibility.

1. Domain Layer (Core Layer)

The **Domain Layer** is the central part of the system and contains the core business logic and entities.

This layer defines the main business models used in the platform such as:

- User
- Property
- Property Image
- Appointment
- Favorite
- Project
- Agent
- Developer

The domain layer does not depend on any external frameworks or technologies. This ensures that the business logic remains independent and reusable.

2. Application Layer

The **Application Layer** contains the business rules and application services that coordinate system operations.

It defines:

- application services
- use case logic
- interfaces for repositories
- business workflows

Examples of responsibilities in this layer include:

- property search logic
- property listing management
- appointment scheduling
- user authentication processes

This layer communicates with the domain layer and defines interfaces that are later implemented in the infrastructure layer.

3. Infrastructure Layer

The **Infrastructure Layer** provides implementations for services defined in the application layer.

This includes:

- database access using **Entity Framework**
- repository implementations
- external service integrations
- data persistence operations

The infrastructure layer communicates with **Microsoft SQL Server** to store and retrieve system data.

4. Presentation Layer (MVC)

The **Presentation Layer** represents the user interface of the Estatefy platform and is implemented using **ASP.NET MVC**.

This layer is responsible for handling user interactions and presenting information through web pages.

It contains three main components:

Models

Represent data structures passed between the controllers and views.

Views

Provide the graphical interface for users using technologies such as:

- HTML
- CSS
- JavaScript

Views display property listings, dashboards, search results, and property details.

Controllers

Controllers process user requests, call application services, and return responses to the views.

Examples include:

- PropertyController
- UserController
- AgentController

- AdminController
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Database Layer

The platform uses **Microsoft SQL Server** as the main database system.

All persistent data such as users, properties, appointments, favorites, and developer projects are stored within the database.

Database interaction is handled using **Entity Framework ORM**, which maps database tables to application models and simplifies data operations.

Benefits of Onion Architecture

Using Onion Architecture provides several important benefits for the Estately platform:

- **Separation of concerns** between system layers
- **Independence of business logic from frameworks**
- **Improved maintainability and scalability**
- **Easier testing of application components**
- **Flexible integration with different technologies**

By combining Onion Architecture with the MVC pattern, the Estately platform achieves a clean and modular design that supports long-term system evolution.